

Exp no:11

Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.

Aim:

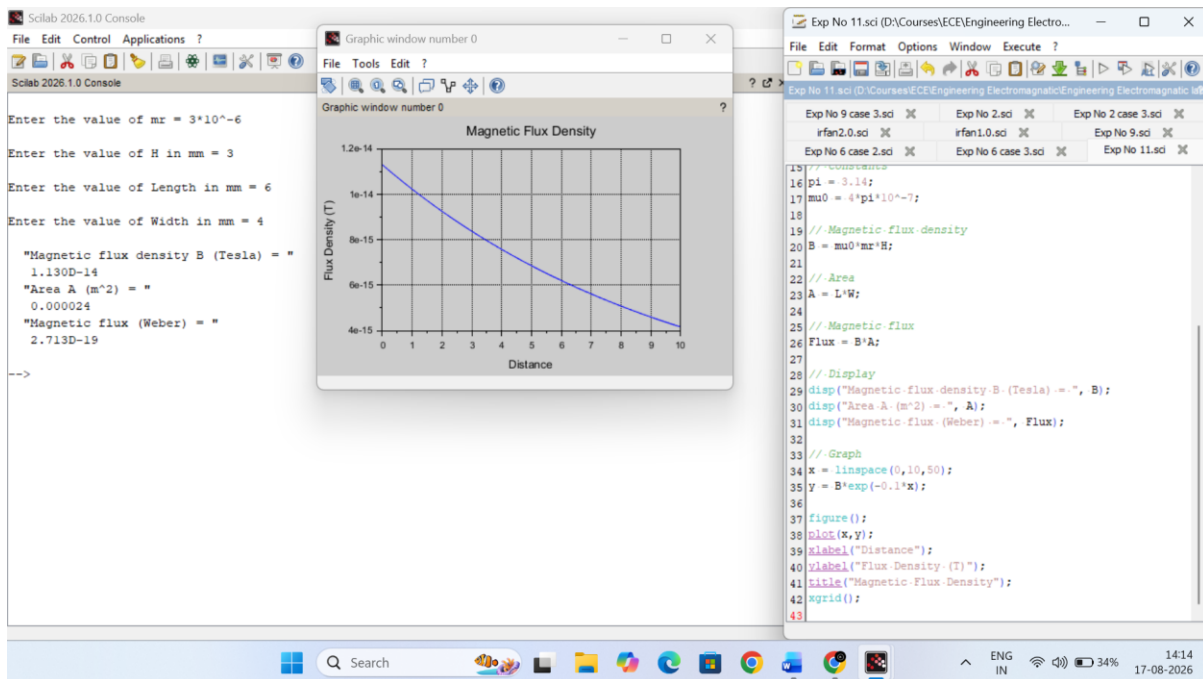
To write program for Magnetic flux density in ferromagnetic materials using SCILAB.

Software required:

- SCILAB version 6.0.1

Theory:

The magnetic flux density (B) in ferromagnetic materials is a measure of the magnetic field's strength within the material, which takes into account both the external magnetic field (H) and the material's inherent magnetic properties (i.e., its magnetization). The relationship between these quantities in a ferromagnetic material is more complex than in non-magnetic or paramagnetic materials because ferromagnetic materials exhibit hysteresis, saturation, and non-linear behavior due to their spontaneous magnetization.



Exp No: 1.1 Determine magnetic flux density in ferromagnetic material

$$\mu_r = 3 \times 10^{-6}$$

$$H = 3 \text{ mm}$$

$$L = 6 \text{ mm}$$

$$W = 4 \text{ mm}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

magnetic flux density

$$B = \mu_0 \mu_r H$$

$$B = (4 \times 3.14 \times 10^{-7}) (3 \times 10^6) (3 \times 10^3)$$

$$B = 1.1304 \times 10^{-4} \text{ T}$$

Area

$$A = L \times W$$

$$= (6 \times 10^{-3}) (4 \times 10^{-3})$$

$$= 24 \times 10^{-6}$$

$$A = 2.4 \times 10^{-5} \text{ m}^2$$

$$I = 0.00024$$

magnetic flux

$$\Phi = B \times A$$

$$= (1.1304 \times 10^{-4}) (2.4 \times 10^{-5})$$

$$\Phi = 2.71296 \times 10^{-9} \text{ Wb}$$

Result :

Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B).

Analytical :

Calculate the magnetic flux density for the following conditions

- The material is ferromagnetic with relative permeability (μ_r) of 500.
- The magnetic field intensity (H) is held constant at 100 A/m.
- The length of the material varies from 0.1 m to 2 m.

Solution:

Magnetic Flux Density in a Ferromagnetic Material:

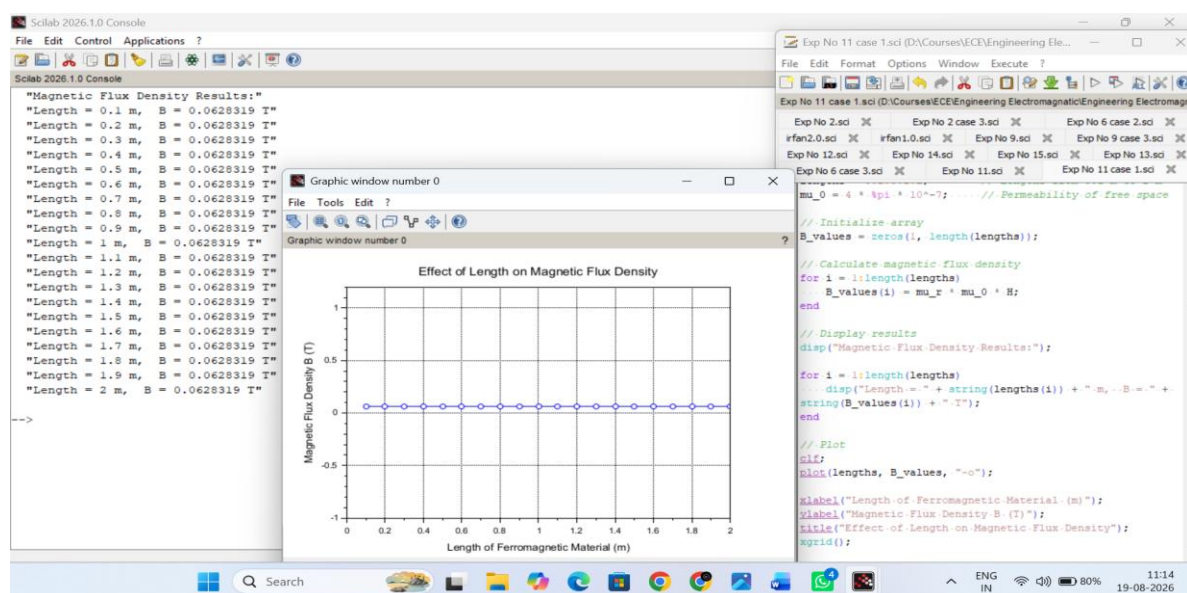
$$B = \mu_0 \mu_r H$$

Substituting the given values:

$$B = (4\pi \times 10^{-7} \text{ T}) \times 500 \times (100 \text{ A/m})$$

$$B = 6.28 \times 10^{-2} \text{ T} = 0.0628 \text{ T}$$

So, for any length of the material between 0.1 m to 2 m, the magnetic flux density B will remain constant at 0.0628 T.



Result: The effect of **length** of a ferromagnetic material on the magnetic flux density (B) is examined.

Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.

Analytical:

Calculate magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity for the following conditions

- **Material:** Soft iron (same as in Case Study 1)
- **Length (L):** Constant at 10 cm
- **Magnetic Field Intensity (H):** Constant at 1000 A/m
- **Width (W):** Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

Solution:

$$B = \mu_0 \mu_r H$$

Substituting the known values:

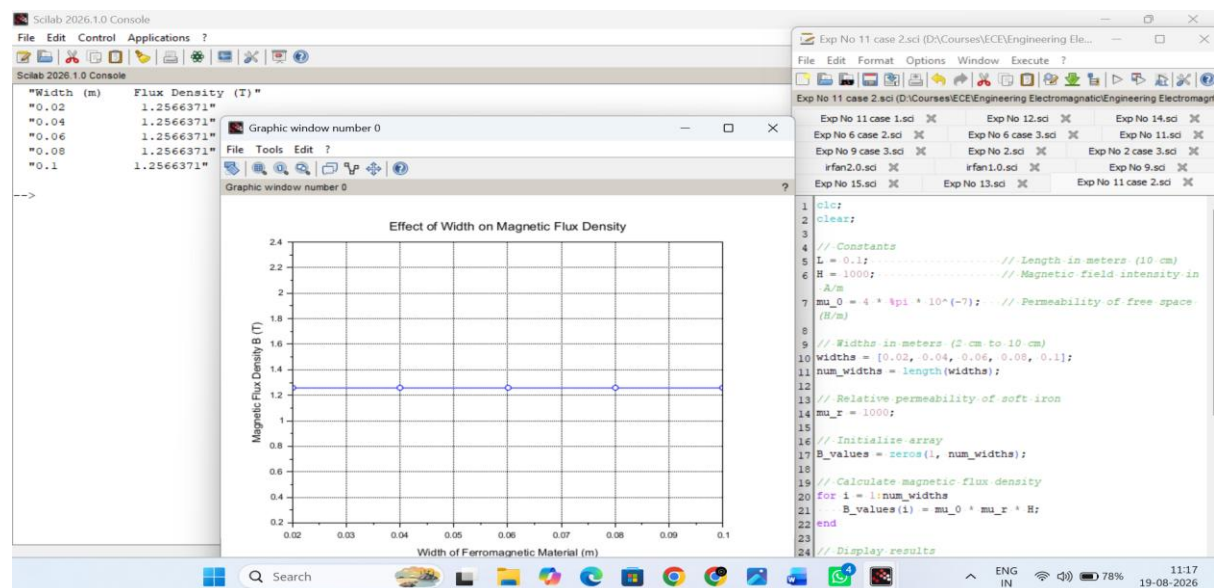
$$B = (4\pi \times 10^{-7}) \times 500 \times 1000$$

$$B = 6.28 \times 10^{-7} \times 500 \times 1000$$

$$B = 0.314 \text{ T}$$

Conclusion:

- The magnetic flux density B **remains constant at 0.314 Tesla** regardless of the width of the material, as long as the length L and magnetic field intensity H remain constant.
- **Width (W)** affects the **total magnetic flux (Φ)** but not the magnetic flux density B.



Result: The effect of **width** of a ferromagnetic material on the magnetic flux density (B) is examined.

Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material change with different values of magnetic field intensity at constant length and width.

Analytical:

Calculate the magnetic flux density (B) in a ferromagnetic material change with different values of magnetic field intensity at constant length and width.

- **Material:** Soft iron (same as in the previous studies)
- **Length (L):** Constant at 10 cm
- **Width (W):** Constant at 5 cm
- **Magnetic Field Intensity (H):** Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)

Solution:

Calculation for Each H:

1. For $H = 200 \text{ A/m}$:

$$B = 4\pi \times 10^{-7} \times 500 \times 200 = 4 \times 3.1416 \times 10^{-7} \times 100000$$

$$B = 2.513 \times 10^{-2} \text{ T} = 0.02513 \text{ T}$$

2. For $H = 500 \text{ A/m}$:

$$B = 4\pi \times 10^{-7} \times 500 \times 500 = 4 \times 3.1416 \times 10^{-7} \times 250000$$

$$B = 6.283 \times 10^{-2} \text{ T} = 0.06283 \text{ T}$$

3. For $H = 1000 \text{ A/m}$:

$$B = 4\pi \times 10^{-7} \times 500 \times 1000 = 4 \times 3.1416 \times 10^{-7} \times 500000$$

$$B = 1.257 \times 10^{-1} \text{ T} = 0.1257 \text{ T}$$

4. For $H = 1500 \text{ A/m}$:

$$B = 4\pi \times 10^{-7} \times 500 \times 1500 = 4 \times 3.1416 \times 10^{-7} \times 750000$$

$$B = 1.884 \times 10^{-1} \text{ T} = 0.1884 \text{ T}$$

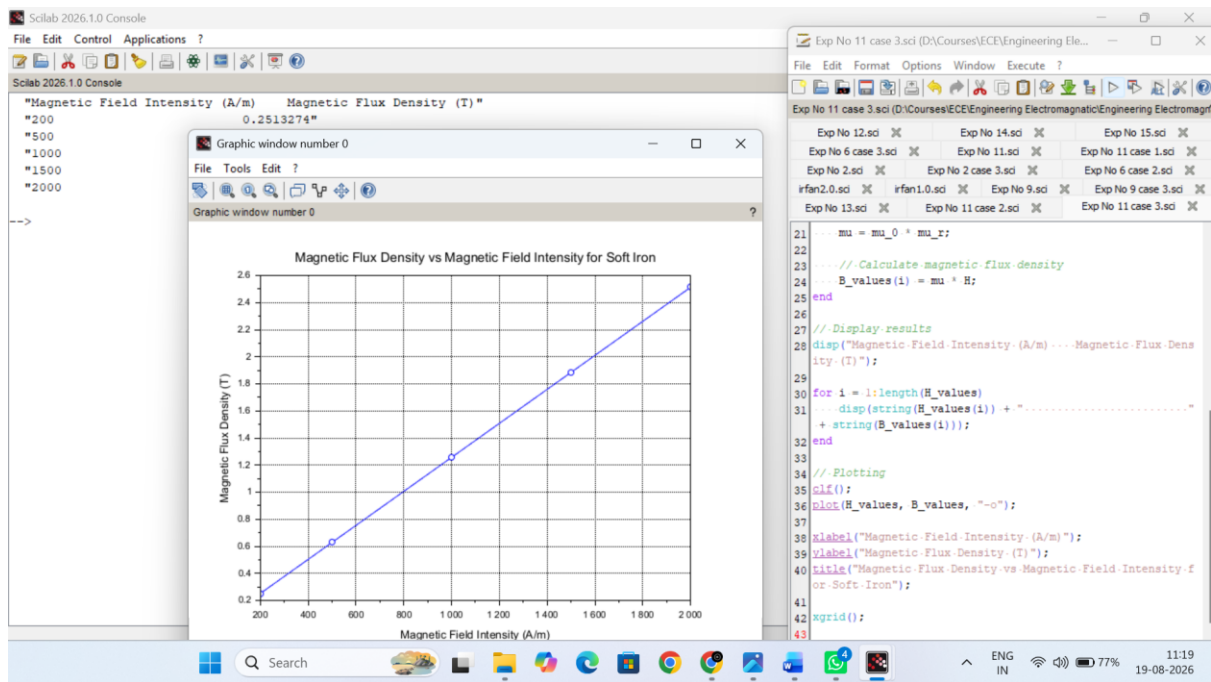
5. For $H = 2000 \text{ A/m}$:

$$B = 4\pi \times 10^{-7} \times 500 \times 2000 = 4 \times 3.1416 \times 10^{-7} \times 1000000$$

$$B = 2.513 \times 10^{-1} \text{ T} = 0.2513 \text{ T}$$

Conclusion:

- The magnetic flux density B **increases** proportionally with the magnetic field intensity H.
- For the given values of H (200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m), the magnetic flux density B increases from **0.02513 T** to **0.2513 T**.
- This behavior follows the linear relationship between B and H, which holds true as long as the material is within the linear region of its magnetization curve (i.e., not saturated).



Result:

The effect of Magnetic field intensity of a ferromagnetic material on the magnetic flux density (B) is examined.